

VAJRA Security Platform - Complete Model Workflow

This document shows the complete database model architecture, relationships, and data flow for all models in the VAJRA Security Platform.

VAJRA Security Platform - Complete Model Workflow Diagram

Database Model Architecture & Data Flow

graph TD

Start([User Action]) --> Frontend[Frontend - Next.js]

Frontend --> StateCheck{State Management}

StateCheck -->|New Action| APIRequest[API Request]

StateCheck -->|Cached Data| DisplayCached[Display Cached Data]

APIRequest --> RouteLayer[FastAPI Router Layer]

RouteLayer --> AuthCheck{Authentication Check}

AuthCheck -->|Invalid| AuthError[Return 401 Unauthorized]

AuthCheck -->|Valid| RouteHandler[Route Handler]

AuthError --> Frontend

%% Database Models Flow

RouteHandler --> ModelSelection{Select Model}

ModelSelection -->|User Operations| UserModel[User Model]

ModelSelection -->|Company Operations| CompanyModel[Company Model]

ModelSelection -->|Threat Operations| ThreatModel[Threat Feed Model]

ModelSelection -->|Alert Operations| AlertModel[Alert Model]

ModelSelection -->|Ransomware Operations| RansomwareModel[Ransomware Incident Model]

ModelSelection -->|Attack Operations| AttackModel[Attack Event Model]

ModelSelection -->|Activity Operations| ActivityModel[Activity Log Model]

%% User Model Flow

UserModel --> UserDB[PostgreSQL - users table]

UserDB --> UserFields{id, email, name, hashed_password, role, is_active, created_at, updated_at}

UserFields --> UserOperations{User Operations}

UserOperations -->|Register| UserRegister[INSERT new user]

UserOperations -->|Login| UserLogin[SELECT user + password verify]

UserOperations -->|Update| UserUpdate[UPDATE user profile]

UserOperations -->|Delete| UserDelete[DELETE user record]

UserRegister --> UserResponse[UserResponse Schema]

UserLogin --> TokenGeneration[JWT Token Generation]

UserUpdate --> UserResponse

UserDelete --> DeleteConfirmation[Delete Confirmation]

TokenGeneration --> TokenStorage[Token Storage]

TokenStorage --> Frontend

%% Company Model Flow

CompanyModel --> CompanyDB[PostgreSQL - companies table]

CompanyDB --> CompanyFields{id, name, domain, industry, description, logo_url, is_active, monitoring_enabled, created_at, updated_at, last_analyzed}

CompanyFields --> CompanyRelations{Relationships}

CompanyRelations --> CompanyThreats[CompanyThreat Model]

CompanyRelations --> CompanyRisk[CompanyRiskAssessment Model]

CompanyThreats --> ThreatDB[PostgreSQL - company_threats table]

ThreatDB --> ThreatFields{id, company_id, threat_type, severity, description, source, confidence_score, status, first_seen, last_seen, created_at, updated_at}

CompanyRisk --> RiskDB[PostgreSQL - company_risk_assessments table]

RiskDB --> RiskFields{id, company_id, risk_level, security_score, active_incidents, abuse_confidence_score, reputation_score, vulnerabilities_count, ssl_valid, domain_age_days, country, isp, assessment_details, created_at, updated_at}

CompanyFields --> CompanyOperations{Company Operations}

CompanyOperations -->|Create| CompanyCreate[INSERT company]

CompanyOperations -->|Read| CompanyRead[SELECT company with relations]

CompanyOperations -->|Update| CompanyUpdate[UPDATE company]

CompanyOperations -->|Delete| CompanyDelete[DELETE company + cascade]

CompanyCreate --> CompanyResponse[Company Schema]

CompanyRead --> CompanyDetailResponse[CompanyDetail Schema]

CompanyUpdate --> CompanyResponse

CompanyDelete --> DeleteConfirmation

%% Threat Feed Model Flow

ThreatModel --> ThreatFeedDB[PostgreSQL - threat_feeds table]

ThreatFeedDB --> ThreatFeedFields{id, feed_name, feed_url, last_updated, status, ioc_count}

ThreatFeedFields --> ThreatOperations{Threat Operations}

ThreatOperations -->|Add Feed| ThreatAdd[INSERT threat feed]

ThreatOperations -->|Update Feed| ThreatUpdate[UPDATE feed status]

ThreatOperations -->|Query Feeds| ThreatQuery[SELECT active feeds]

ThreatAdd --> ThreatResponse[ThreatFeed Schema]

ThreatUpdate --> ThreatResponse

ThreatQuery --> ThreatListResponse[List[ThreatFeed Schema]]

%% Alert Model Flow

AlertModel --> AlertDB[PostgreSQL - alerts table]

AlertDB --> AlertFields{id, title, severity, description, source, time, created_at}

AlertFields --> AlertOperations{Alert Operations}

AlertOperations -->|Create| AlertCreate[INSERT alert]

AlertOperations -->|Read| AlertRead[SELECT alerts]

AlertOperations -->|Update| AlertUpdate[UPDATE alert status]

AlertOperations -->|Delete| AlertDelete[DELETE alert]

AlertCreate --> AlertResponse[Alert Schema]

AlertRead --> AlertListResponse[List[Alert Schema]]

AlertUpdate --> AlertResponse

AlertDelete --> DeleteConfirmation

%% Ransomware Model Flow

RansomwareModel --> RansomwareDB[PostgreSQL - ransomware_incidents table]

RansomwareDB --> RansomwareFields{id, group_name, target, country, published_date, impact, status, description, created_at}

RansomwareFields --> RansomwareOperations{Ransomware Operations}

RansomwareOperations -->|Add Incident| RansomwareAdd[INSERT incident]
RansomwareOperations -->|Query| RansomwareQuery[SELECT incidents]
RansomwareOperations -->|Group Query| RansomwareGroupQuery[SELECT by group_name]

RansomwareAdd --> RansomwareResponse[RansomwareIncident Schema]
RansomwareQuery --> RansomwareListResponse[List[RansomwareIncident Schema]]
RansomwareGroupQuery --> RansomwareGroupResponse[List[RansomwareIncident Schema]]

%% Attack Event Model Flow

AttackModel --> AttackDB[PostgreSQL - attack_events table]
AttackDB --> AttackFields{id, event_type, source_ip, target_ip, source_country, target_country, attack_vector, severity, description, timestamp}
AttackFields --> AttackOperations{Attack Operations}

AttackOperations -->|Log Event| AttackLog[INSERT attack event]
AttackOperations -->|Query| AttackQuery[SELECT events]
AttackOperations -->|Map Query| AttackMapQuery[SELECT for attack map]

AttackLog --> AttackResponse[AttackEvent Schema]
AttackQuery --> AttackListResponse[List[AttackEvent Schema]]
AttackMapQuery --> AttackMapResponse[Attack event map data]

%% Activity Log Model Flow

ActivityModel --> ActivityDB[PostgreSQL - activity_logs table]
ActivityDB --> ActivityFields{id, user_id, action, resource, details, ip_address, timestamp}
ActivityFields --> ActivityOperations{Activity Operations}

ActivityOperations -->|Log Activity| ActivityLogInsert[INSERT activity log]
ActivityOperations -->|Query| ActivityQuery[SELECT user activities]
ActivityOperations -->|Audit| ActivityAudit[SELECT for audit trail]

ActivityLogInsert --> ActivityResponse[ActivityLog Schema]
ActivityQuery --> ActivityListResponse[List[ActivityLog Schema]]
ActivityAudit --> ActivityAuditResponse[Audit trail data]

%% Service Layer Integration

UserResponse --> ServiceLayer[Service Layer]
CompanyDetailResponse --> ServiceLayer
ThreatListResponse --> ServiceLayer

AlertListResponse --> ServiceLayer

RansomwareListResponse --> ServiceLayer

AttackMapResponse --> ServiceLayer

ActivityAuditResponse --> ServiceLayer

ServiceLayer --> ExternalServices{External Services}

ExternalServices --> AlienVault[AlienVault OTX Service]

ExternalServices --> RansomwareLive[Ransomware.live Service]

ExternalServices --> HackerNews[Hacker News Service]

ExternalServices --> VirusTotal[VirusTotal Service]

ExternalServices --> AbuseIPDB[AbuseIPDB Service]

ExternalServices --> SSL Labs[SSL Labs Service]

ExternalServices --> DomainAnalysis[Domain Analysis Service]

ExternalServices --> Gemini[Gemini AI Service]

ExternalServices --> HuggingFace[HuggingFace Service]

AlienVault --> AlienVaultAPI[AlienVault OTX API]

RansomwareLive --> RansomwareLiveAPI[Ransomware.live API]

HackerNews --> HackerNewsAPI[Hacker News API]

VirusTotal --> VirusTotalAPI[VirusTotal API]

AbuseIPDB --> AbuseIPDBAPI[AbuseIPDB API]

SSL Labs --> SSL LabsAPI[SSL Labs API]

DomainAnalysis --> DomainAnalysisAPI[Domain Analysis APIs]

Gemini --> GeminiAPI[Gemini AI API]

HuggingFace --> HuggingFaceAPI[HuggingFace API]

%% Data Processing Flow

AlienVaultAPI --> DataProcessing[Data Processing]

RansomwareLiveAPI --> DataProcessing

HackerNewsAPI --> DataProcessing

VirusTotalAPI --> DataProcessing

AbuseIPDBAPI --> DataProcessing

SSL LabsAPI --> DataProcessing

DomainAnalysisAPI --> DataProcessing

GeminiAPI --> DataProcessing

HuggingFaceAPI --> DataProcessing

DataProcessing --> DataTransformation{Data Transformation}

DataTransformation --> DataValidation[Data Validation]

DataValidation --> DataStorage[Database Storage]

DataValidation --> DataCache[Redis Cache]

DataStorage --> ServiceLayer

DataCache --> ServiceLayer

%% Response Flow

ServiceLayer --> ResponseFormatting[Response Formatting]

ResponseFormatting --> SchemaValidation[Schema Validation]

SchemaValidation --> APIResponse[API Response]

APIResponse --> Frontend

Frontend --> StateUpdate[State Update]

StateUpdate --> ComponentRender[Component Render]

ComponentRender --> DisplayResult[Display Result]

DisplayResult --> WebSocketNotify{WebSocket Notification}

WebSocketNotify -->|Real-time Update| WebSocketPush[Push Update via WebSocket]

WebSocketPush --> OtherClients[Other Clients]

DisplayResult --> End([User Action Complete])

%% Styling

style Start fill:#4CAF50,color:#fff

style End fill:#f44336,color:#fff

style Frontend fill:#2196F3,color:#fff

style UserModel fill:#9C27B0,color:#fff

style CompanyModel fill:#FF9800,color:#fff

style ThreatModel fill:#4CAF50,color:#fff

style AlertModel fill:#f44336,color:#fff

style RansomwareModel fill:#E91E63,color:#fff

style AttackModel fill:#00BCD4,color:#fff

style ActivityModel fill:#607D8B,color:#fff

style ServiceLayer fill:#FFC107,color:#000

style ExternalServices fill:#8BC34A,color:#000

style DataProcessing fill:#03A9F4,color:#fff

Detailed Model Relationships

User Model Relationships

User Model (users table)

- Fields: id, email, name, hashed_password, role, is_active, created_at, updated_at
- Relationships: None (standalone model)
- Operations: Register, Login, Update Profile, Delete
- Security: Password hashing, JWT token generation

Company Model Relationships

Company Model (companies table)

- Fields: id, name, domain, industry, description, logo_url, is_active, monitoring_enabled, created_at, updated_at, last_analyzed
- Relationships:
 - One-to-Many with CompanyThreat (company_id)
 - One-to-Many with CompanyRiskAssessment (company_id)
- Operations: Create, Read, Update, Delete with cascade
- Monitoring: Continuous monitoring status tracking

CompanyThreat Model Relationships

CompanyThreat Model (company_threats table)

- Fields: id, company_id, threat_type, severity, description, source, confidence_score, status, first_seen, last_seen, created_at, updated_at
- Relationships:
 - Many-to-One with Company (company_id)
- Operations: Create, Read, Update, Delete
- Tracking: Threat lifecycle management (ACTIVE, RESOLVED, IGNORED)

CompanyRiskAssessment Model Relationships

CompanyRiskAssessment Model (company_risk_assessments table)

- Fields: id, company_id, risk_level, security_score, active_incidents, abuse_confidence_score, reputation_score, vulnerabilities_count, ssl_valid, domain_age_days, country, isp, assessment_details, created_at, updated_at
- Relationships:
 - Many-to-One with Company (company_id)
- Operations: Create, Read, Update, Delete
- Assessment: Comprehensive risk scoring and tracking

Alert Model Relationships

Alert Model (alerts table)

- Fields: id, title, severity, description, source, time, created_at
- Relationships: None (standalone model)
- Operations: Create, Read, Update, Delete
- Severity Levels: critical, high, medium, low

ThreatFeed Model Relationships

ThreatFeed Model (threat_feeds table)

- Fields: id, feed_name, feed_url, last_updated, status, ioc_count
- Relationships: None (standalone model)
- Operations: Add Feed, Update Feed, Query Feeds
- Status Tracking: active, inactive, error

RansomwareIncident Model Relationships

RansomwareIncident Model (ransomware_incidents table)

- Fields: id, group_name, target, country, published_date, impact, status, description, created_at
- Relationships: None (standalone model)
- Operations: Add Incident, Query Incidents, Group Query
- Impact Levels: Critical, High, Medium, Low

AttackEvent Model Relationships

AttackEvent Model (attack_events table)

- Fields: id, event_type, source_ip, target_ip, source_country, target_country, attack_vector, severity, description, timestamp
- Relationships: None (standalone model)
- Operations: Log Event, Query Events, Map Query
- Geographic: Source and target country tracking

ActivityLog Model Relationships

ActivityLog Model (activity_logs table)

- Fields: id, user_id, action, resource, details, ip_address, timestamp
- Relationships: None (standalone model)
- Operations: Log Activity, Query Activities, Audit Trail
- Audit: Complete user action tracking

Service Layer Data Flow

AlienVault Service Flow

Request → AlienVault Service → OTX API → Threat Data → IOC Processing → Database Storage
→ Response

Ransomware Service Flow

Request → Ransomware Service → Ransomware.live API → Incident Data → Statistics Calculation → Database Storage → Response

Hacker News Service Flow

Request → Hacker News Service → HN API → News Stories → Keyword Filtering → Relevance Scoring → Response

Domain Analysis Service Flow

Request → Domain Analysis Service → Multiple APIs → Risk Assessment → SSL Check → Reputation Check → Response

PDF Service Flow

Request → PDF Service → Data Aggregation → Template Processing → Black/White Styling → PDF Generation → File Download

API Route to Model Mapping

Authentication Routes

POST /api/auth/register → User Model (INSERT)

POST /api/auth/login → User Model (SELECT) + JWT Generation

POST /api/auth/refresh → User Model (SELECT) + Token Refresh

Dashboard Routes

GET /api/dashboard/summary → Multiple Models + External APIs

GET /api/dashboard/alerts → Alert Model (SELECT)

GET /api/dashboard/attack-map → AttackEvent Model (SELECT)

Company Routes

GET /api/companies → Company Model (SELECT)

GET /api/companies/{id} → Company Model + Relations (SELECT)

POST /api/companies → Company Model (INSERT)

PUT /api/companies/{id} → Company Model (UPDATE)

DELETE /api/companies/{id} → Company Model (DELETE with cascade)

Threat Intelligence Routes

GET /api/threat-intelligence → ThreatFeed Model + AlienVault Service

GET /api/threat-intelligence/trend → Historical Data Processing

GET /api/threat-intelligence/actors → Threat Actor Data

GET /api/threat-intelligence/industries → Industry Data

Ransomware Routes

GET /api/ransomware → RansomwareIncident Model + Ransomware Service

GET /api/ransomware/stats → Statistics Calculation

GET /api/ransomware/group/{name} → RansomwareIncident Model (WHERE group_name)

Alert Routes

GET /api/alerts → Alert Model (SELECT)

POST /api/alerts → Alert Model (INSERT)

PUT /api/alerts/{id} → Alert Model (UPDATE)

DELETE /api/alerts/{id} → Alert Model (DELETE)

Report Routes

GET /api/reports/threat-intelligence → PDF Service + Threat Data

GET /api/reports/ransomware → PDF Service + Ransomware Data

GET /api/reports/global-attacks → PDF Service + Attack Data

GET /api/reports/company/{id} → PDF Service + Company Data

GET /api/reports/executive → PDF Service + Executive Data

GET /api/reports/comprehensive → PDF Service + All Data

Data Validation Flow

User Input → Frontend Validation → API Request → Schema Validation → Database Validation →
Business Logic Validation → Response

Error Handling Flow

Error Detection → Error Logging → User Notification → Fallback Data → Graceful Degradation → Recovery Attempt

Caching Strategy Flow

Request → Cache Check → Cache Hit → Return Cached Data

Request → Cache Check → Cache Miss → Database Query → Cache Update → Return Data

WebSocket Real-time Flow

Event Trigger → WebSocket Manager → Connected Clients → Event Broadcast → Client Update → UI Refresh

Background Job Flow

Scheduler → Job Trigger → External API Call → Data Processing → Database Update → Cache Refresh → WebSocket Notification

Security Flow

Request → JWT Validation → Permission Check → Rate Limiting → Input Sanitization → SQL Injection Prevention → XSS Prevention → Processing → Response

Database Model Structures

User Model Structure

Field	Type	Description
id	Integer	Primary Key
email	String	Unique, Indexed
name	String	User Name
hashed_password	String	Hashed Password
role	String	SOC Analyst (default)
is_active	Boolean	Active Status
created_at	DateTime	Creation Timestamp
updated_at	DateTime	Update Timestamp

Company Model Structure

Field	Type	Description
id	Integer	Primary Key
name	String	Company Name
domain	String	Unique Domain
industry	String	Industry Type
description	Text	Company Description
logo_url	String	Logo URL
is_active	Boolean	Active Status
monitoring_enabled	Boolean	Monitoring Status
created_at	DateTime	Creation Timestamp
updated_at	DateTime	Update Timestamp
last_analyzed	DateTime	Last Analysis

CompanyThreat Model Structure

Field	Type	Description
id	Integer	Primary Key
company_id	Integer	Foreign Key (companies)
threat_type	String	Threat Type
severity	String	LOW/MEDIUM/HIGH/CRITICAL
description	Text	Threat Description
source	String	Threat Source
confidence_score	Integer	0-100 Confidence

status	String	ACTIVE/RESOLVED/IGNORED
first_seen	DateTime	First Detection
last_seen	DateTime	Last Detection
created_at	DateTime	Creation Timestamp
updated_at	DateTime	Update Timestamp

CompanyRiskAssessment Model Structure

Field	Type	Description
id	Integer	Primary Key
company_id	Integer	Foreign Key (companies)
risk_level	String	LOW/MEDIUM/HIGH/CRITICAL
security_score	Integer	0-100 Score
active_incidents	Integer	Active Incident Count
abuse_confidence_score	Integer	Abuse Confidence
reputation_score	Integer	Reputation Score
vulnerabilities_count	Integer	Vulnerability Count
ssl_valid	Boolean	SSL Certificate Status
domain_age_days	Integer	Domain Age in Days
country	String	Country
isp	String	ISP/Provider
assessment_details	Text	Additional Details (JSON)
created_at	DateTime	Creation Timestamp
updated_at	DateTime	Update Timestamp

Alert Model Structure

Field	Type	Description
id	Integer	Primary Key
title	String	Alert Title
severity	String	critical/high/medium/low
description	Text	Alert Description
source	String	Alert Source
time	String	Alert Time
created_at	DateTime	Creation Timestamp

RansomwareIncident Model Structure

Field	Type	Description
id	Integer	Primary Key
group_name	String	Ransomware Group

target	String	Target Organization
country	String	Target Country
published_date	DateTime	Publication Date
impact	String	Critical/High/Medium/Low
status	String	Published Status
description	Text	Incident Description
created_at	DateTime	Creation Timestamp

AttackEvent Model Structure

Field	Type	Description
id	Integer	Primary Key
event_type	String	Attack Event Type
source_ip	String	Source IP Address
target_ip	String	Target IP Address
source_country	String	Source Country
target_country	String	Target Country
attack_vector	String	Attack Vector
severity	String	Severity Level
description	Text	Event Description
timestamp	DateTime	Event Timestamp

ActivityLog Model Structure

Field	Type	Description
id	Integer	Primary Key
user_id	Integer	User ID
action	String	User Action
resource	String	Affected Resource
details	Text	Action Details
ip_address	String	IP Address
timestamp	DateTime	Action Timestamp